Micro Hack 1 — Business Apps

Participant Workbook · End-customer Apps (Rayfin)

FY27 EMEA EPS — Fabric Apps Motion

Contents

Micro Hack 1 — Business Apps Workbook (Participants)

Track: **End-customer Apps (Rayfin)** Scenario: **Helios Bicycle** (fictional final customer)

> Print-ready PDF: `microhack-1-business-apps-workbook.pdf`

This is a **didactic** micro-hack. You don't need prior Rayfin experience — every step explains **what** you do, **why**, and **what you should see**. Follow them in order.

Day agenda (1 day)

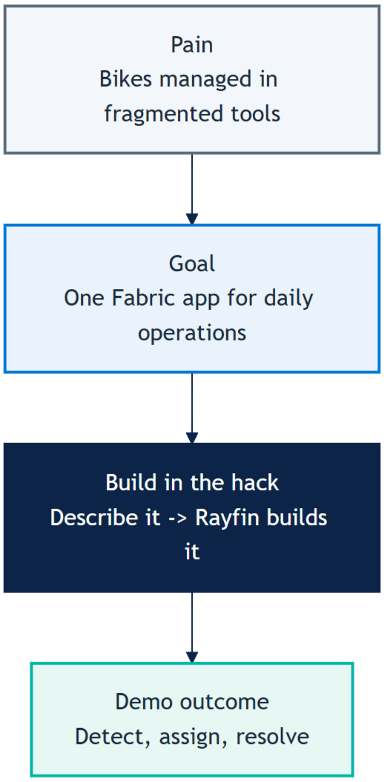
Morning — presentations & demos. Afternoon — hands-on hack. This is a **standalone one-day micro-hack for the Business Apps (Rayfin) track**.

Time	Session
09:00	Welcome & motion overview — the Fabric apps motion for EMEA
09:30	Microsoft Fabric foundations — OneLake, capacity, workspaces
10:00	Rayfin foundations — spec-driven apps, data model = database
10:30	Break
10:45	**Demo · Helios Bicycle Studio** — the app you'll build today
11:15	The hack brief, teams & environment check
12:00	Lunch
13:30	**Hack · Sprint 1** — build the core app (steps 1–5)
15:30	Break
15:45	**Hack · Sprint 2** — extend, polish & deploy (steps 6–9)
16:45	Team demos (5 min each)
17:00	Wrap-up, KPIs & next steps

Part A — Understand

1) The scenario in plain words

Helios Bicycle runs shared city bikes across several stations. Today the station teams track bikes, repairs and rider feedback in spreadsheets and chat — slow and error-prone. Your job in the afternoon: build **one simple app**, *Helios Bicycle Studio*, that lets a manager see every bike, fix the right one first, and keep riders happy.



2) What is Rayfin? (60-second primer)

Rayfin is a *Backend-as-a-Service on Microsoft Fabric*. In plain terms:

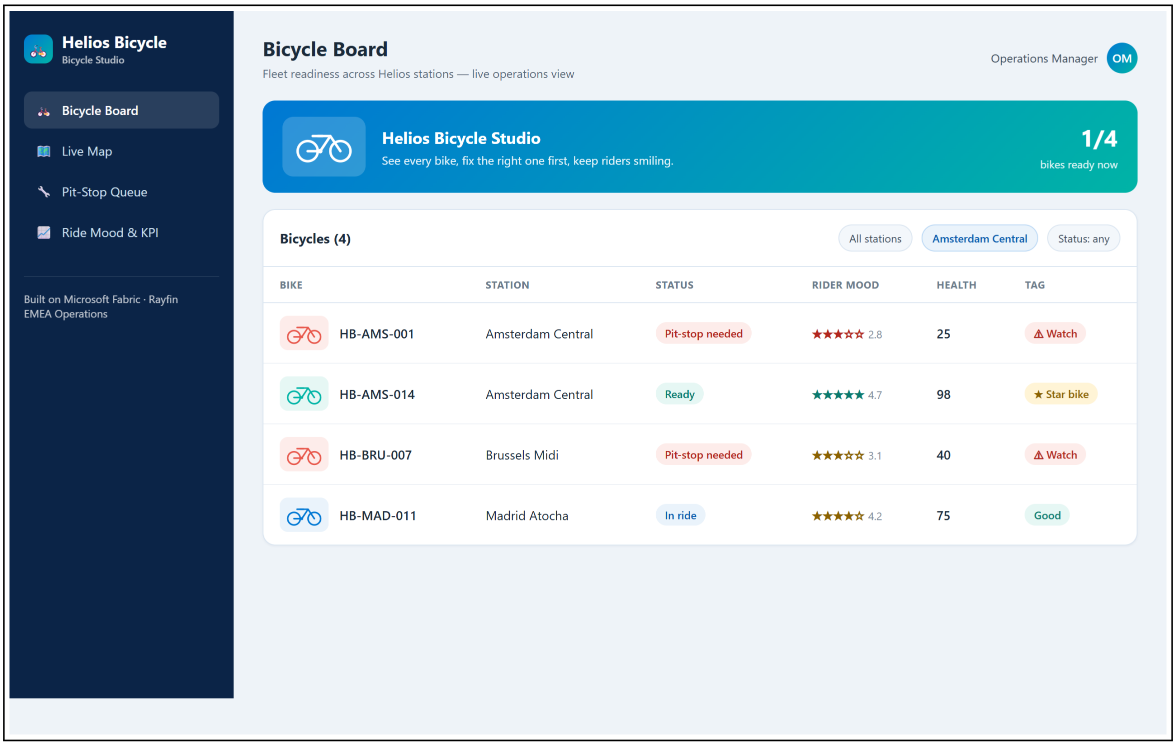
- You **describe your data** (e.g. a `Bicycle` has a code, a station, a status). Rayfin then **creates and manages the database** for you in Fabric — no SQL, no servers to run.

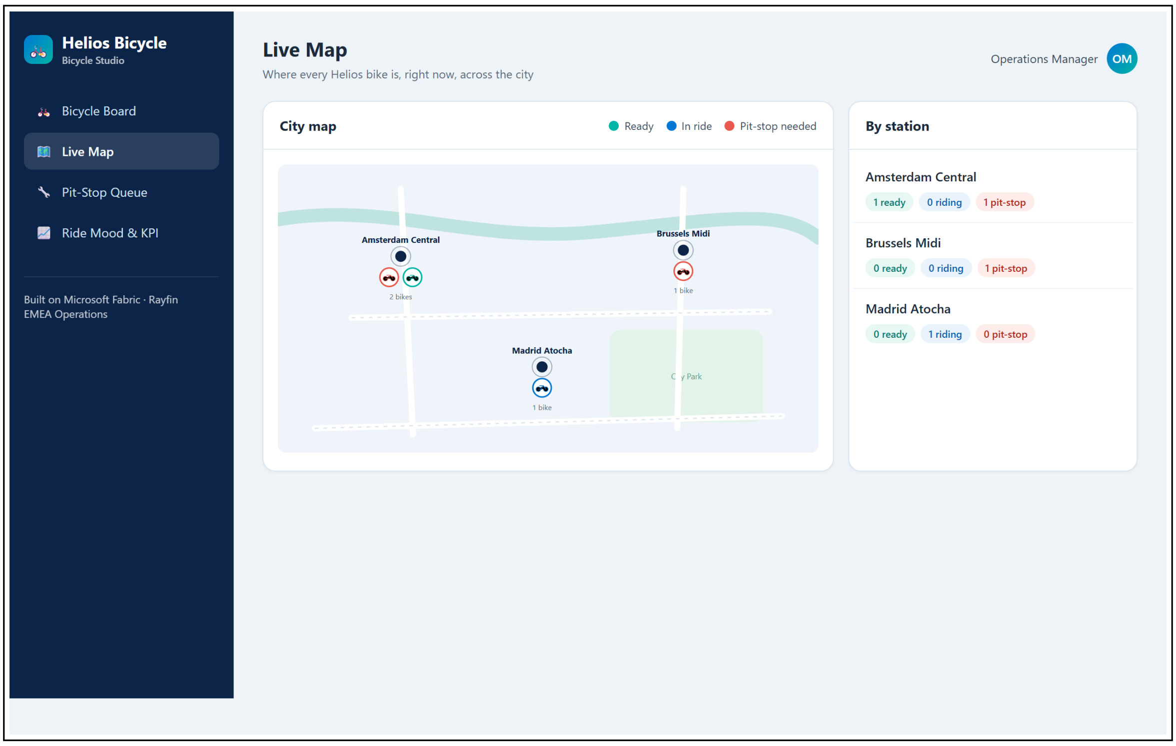
- You **describe your screens in natural language** (with GitHub Copilot). Rayfin generates the app UI against that data.
- So building an app becomes: *declare the data -> describe the screens -> run*.

Key idea to remember: **your data model = your database**. When you declare a `Bicycle` type, Rayfin makes the matching table automatically.

3) The result you are aiming for

By the end of the afternoon your app should look like this — a **Bicycle Board** and a **Live Map** of the fleet:





Step 0 — Prerequisites & environment setup (do this first)

Modeled on Microsoft's Build LAB514 setup. On a lab VM most tools are pre-installed.

Tools you need:

- Node.js 20+ (24 recommended) and npm
- Visual Studio Code
- **GitHub Copilot CLI** (the `copilot` terminal agent)

Sign in (two accounts):

1. **GitHub Copilot** — in a terminal run `copilot`, then `/login`, choose **GitHub.com**, finish the browser sign-in, confirm *"Signed in successfully"*, then `/exit`.
2. **Microsoft Fabric** — open `https://app.fabric.microsoft.com` and sign in. *(If it shows "Power BI" bottom-left, switch it to "Fabric".)*

Tenant settings (ask your Fabric Administrator to enable these first). In **-> Admin portal -> Tenant settings**, search and enable both, clicking **Apply** each time:

- **"Users can create Fabric items"** — search *"Fabric items"* -> enable -> **Apply**.
- **"Users can discover and create org apps (preview)"** (Fabric Apps) — search *"org apps"* -> enable -> **Apply**.

  Without these two tenant settings, deploying a Rayfin app to Fabric fails. They are **tenant-admin** settings — a non-admin participant cannot turn them on themselves.

Fabric workspace & capacity: create (or use) a workspace assigned to a **Fabric capacity** that supports these items — this is where your app deploys. A capacity-less (Pro-only) workspace will not host the app.

Verify your setup:

```
node --version
npm --version
copilot --version
```

First deploy uses `rayfin login` + `rayfin up`. Once you have scaffolded the project (Step 1), the **first time** you bring it up, run these two explicitly before relying on the `npm run dev` loop — they authenticate you against Fabric (Entra) and create the app item:

```
npx rayfin login     # interactive Entra sign-in (also fixes later 401/403)
npx rayfin up       # first full deploy: creates the Fabric item + provisions the DB
```

> **You must test from inside Fabric.** A Rayfin app is connected to a Fabric semantic model, so opening the raw app URL in a normal browser tab shows **"Can't open this app outside Fabric — Opening apps connected to semantic models outside of the Fabric portal is not supported at this time."** Always open and validate the app **from the Fabric portal** (`app.fabric.microsoft.com` -> your workspace -> the app item), not from the bare URL.

What you should see. All three commands print a version, you're signed in to GitHub Copilot and to Fabric (capacity-backed workspace, both tenant settings enabled), and `npx rayfin up` prints a live Fabric app link.

Part B — Build it step by step

> **How to work. There is no repository to clone** for this track — Rayfin scaffolds the whole project for you with one command (Step 1). After that you build the app **conversationally** with the **GitHub Copilot CLI** (`copilot` in your terminal): you give it a short prompt, it writes the code, and you check the result in the browser. Each step below = one action + **what you should see**.

Who does what — Copilot vs. the Rayfin CLI (read once)

This is the key thing to understand: **the prompt + Copilot do NOT deploy anything by themselves.**

- **GitHub Copilot** = writes/edits your TypeScript: the data model in `rayfin/data/` and the screens. It does **not** touch Fabric or your database.

- **The Rayfin CLI** = *you* run it. It deploys to Fabric, **provisions the database from your data model**, and hosts the app. `npm run dev` wraps this for the dev loop.

So the rhythm is: *ask Copilot to write code -> run a CLI command to deploy/provision -> look at the app.*

Command	What it does
`npm run dev`	Dev loop: deploys (`rayfin up`) + serves the app for fast iteration. **Use this most of the time.**
`npx rayfin up`	Full deploy to Fabric: creates the app item, applies the DB schema from your data model, builds & uploads the UI, prints the live URL.
`npx rayfin up db apply`	After you change `rayfin/data/`, push only the new schema (add `--force` if it warns about data loss).
`npx rayfin up staticapp deploy`	Redeploy only the frontend (faster).
`npx rayfin up status`	Show the current deployment.
`npx rayfin up --dry-run`	Preview a deploy without changing anything.
`npx rayfin login --tenant <tenant-id>`	Re-authenticate if you hit a 401/403 (scope to your Entra tenant).

> **Auth note.** Email/password sign-in only works in local dev. Once deployed to Fabric, only **Fabric brokered auth (Entra SSO)** works — `rayfin.yml` must have `auth.fabric.enabled: true`.

The build loop (what happens between prompts)

Every build step follows the same rhythm:

1. **You** give the GitHub Copilot CLI a short prompt.
2. **Copilot** writes/edits the code **and validates it for you** — it type-checks (`tsc`), lints (`eslint`), builds (`npm run build`) and runs the unit tests, fixing issues until green.
3. **You** run `npm run dev` to see the change in the browser (it re-deploys and re-provisions the database if the data model changed).
4. **You** sanity-check the screen, then move to the next prompt.

> If Copilot reports build/test errors it couldn't fix on its own, paste the error back to it — that's the normal loop. Don't move on while the build is red.

Step 1 — Create the project (no clone needed)

What you do. Scaffold a Rayfin project. "Scaffold" means *generate a ready-to-run starter project* — you do **not** clone any repo.

```
npm create @microsoft/rayfin@latest --template dataapp
cd <the-folder-it-created>
```

> **Pick a valid template.** `dataapp` is the recommended base (a Fabric-authenticated React + Vite app wired for Rayfin data — you add entities in `rayfin/data/`). Running `npm create @microsoft/rayfin@latest` with no `--template` lets you choose interactively from `blankapp`, `dataapp`, `gettingstartedauth`, `todoapp`.

Why. Rayfin gives you a complete app skeleton (data layer + UI) so you start from something that already runs.

What you should see. A new project folder is created and dependencies install.

Step 2 — Run the empty app first (your "it works" check)

What you do. Start the starter app **before changing anything** — exactly like a Hello World, to confirm your environment and Fabric connection are healthy.

```
npx rayfin login      # first time only (or on a 401/403) — interactive Entra sign-in
npx rayfin up        # first full deploy: creates the Fabric item + provisions the DB
npm run dev           # dev loop: re-deploys + serves the app for fast iteration
```

Why. The first `npx rayfin up` authenticates you and **creates the Fabric app item**, then provisions your database from the data model in `rayfin/data/*.ts`. `npm run dev` wraps `rayfin up` for the fast iteration loop, but running `login` + `up` once up front makes the first deploy reliable. If the starter app opens, your whole chain works and you can safely start customizing.

What you should see. `npx rayfin up` prints a **live Fabric app link**; open it **from the Fabric portal** and the starter app loads, with the new app showing in your workspace. **Don't go further until this works.**

> **Open it from Fabric, not the bare URL.** If you paste the raw app URL into a normal browser tab you'll get *"Can't open this app outside Fabric — Opening apps connected to semantic models outside of the Fabric portal is not supported at this time."* That's expected — always validate from `app.fabric.microsoft.com` -> your workspace -> the app item.

Step 3 — Declare the data model

What you do. Tell Copilot the data your app is about. This defines the **database tables**.

```
Build "Helios Bicycle Studio", a fun bike-operations app on Rayfin.
Data model:
- Bicycle(bikeCode, station, status[ready|in-ride|pit-stop-needed], featured)
- RideSession(bicycle, riderAlias, moodScore, startedAt, endedAt)
- PitStopTicket(ticketCode, bicycle, station, priority[high|normal],
                status[new|assigned|done], issue, openedAt, assignedMechanic)
- MechanicProfile(displayName, station, active)
Roles: Operations Manager has full access; Mechanic sees only assigned tickets.
Style: clean, friendly, Microsoft Fluent, blue #0078d4 + teal #00b4a6.
```

Why. These four types become four tables. `Bicycle.status` is a fixed list of values, so the app can colour bikes by status later.

What you should see. Copilot creates `rayfin/data/*.ts` — one `@entity()` class per type, marked `@authenticated('*')`, with relationships (e.g. a `@one` link from `PitStopTicket` to `Bicycle`, and an **optional/nullable** `assignedMechanic`) — exports them from `schema.ts`, and enables the data service in `rayfin.yml` (`dialect: mssql`).

Then — apply the schema. Re-run `npm run dev` (or `npx rayfin up db apply`) so Rayfin provisions the matching tables. The screens are still empty — that's expected.

> **Roles are app-level — important.** Rayfin's data layer only knows `authenticated` / anonymous; there are no per-row database roles. So the **Operations Manager vs Mechanic** split is enforced **in the app** (a role context + a header switcher; the Mechanic gets a focused "My Pit-Stops" worklist). When you ask Copilot for roles in Step 8, expect this app-layer logic — not database roles.
>
> **Load demo data — your DB starts empty.** Before the screens can show anything, add data. Ask Copilot for a **Manager-only "Load demo fleet" button** that writes real rows through the data API (no hardcoded display data), then click it once:
>
> ```
> Add a Manager-only "Load demo fleet" button that seeds the database with a
>  ↪  few bikes,
> mechanics, ride sessions and pit-stop tickets via the Rayfin data API (no
>  ↪  hardcoded UI data).
> ```

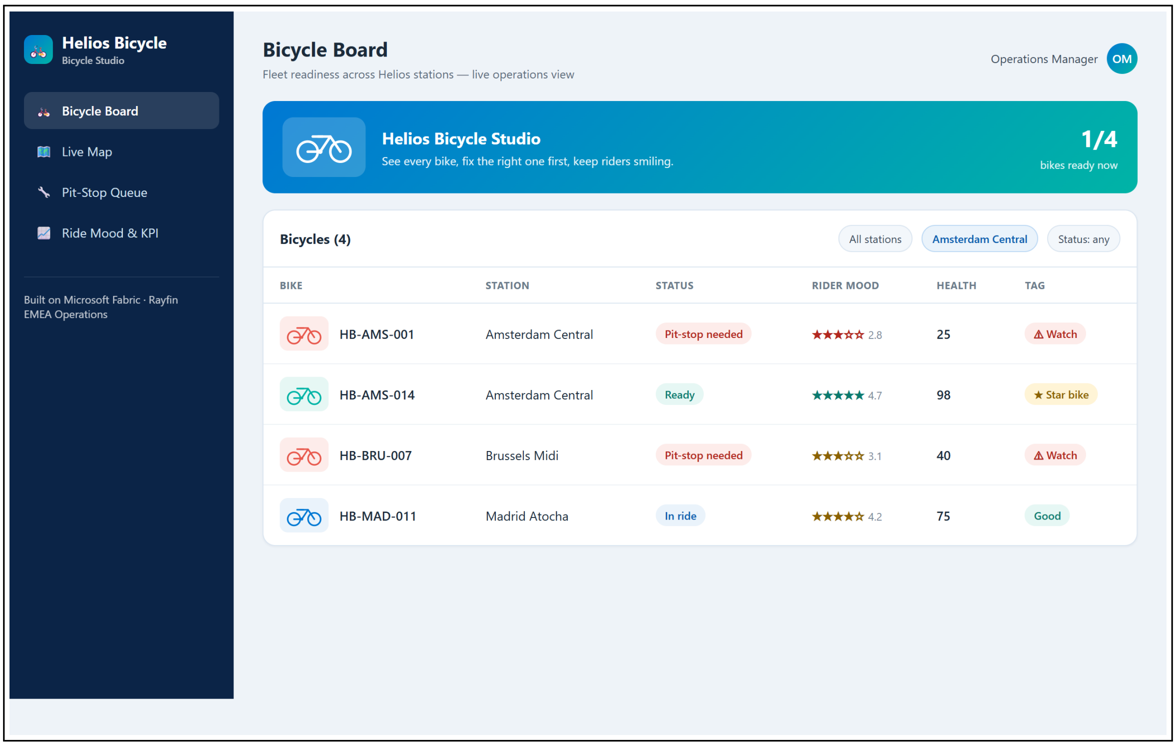
Step 4 — Build the Bicycle Board (your first screen)

What you do. Ask for the main screen: the list of bikes a manager looks at every morning.

```
Add a "Bicycle Board" screen: a table grouped by station showing a status pill,
rider mood as stars, a health score and a tag (Star / Good / Watch).
Add filters for station and status.
```

Why. This is the app's home base — *detect* which bikes need attention. The health score/tag is computed from status + mood so the manager sees priorities at a glance.

What you should see. A table of bikes by station with coloured status pills and a tag. You can filter by station and status.



Now deploy & check it in Fabric. Push this change live with **Step 9** and open the app **from the Fabric portal**. Since the app is already deployed, do a **clean redeploy every time**: first run **Annex A** (delete the item in Fabric -> clear local state -> `npx rayfin up`) so the next deploy doesn't fail with a **404**.

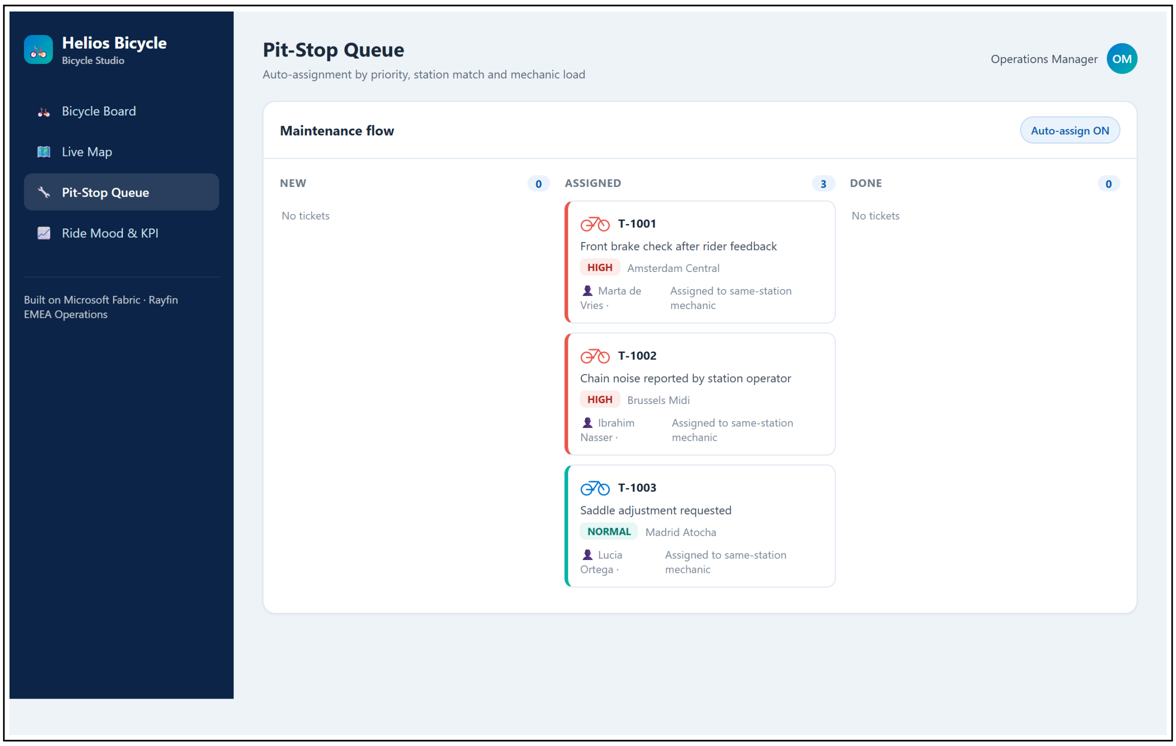
Step 5 — Build the Pit-Stop Queue (take action)

What you do. Add the screen where a manager turns a problem into an assigned repair.

```
Add a "Pit-Stop Queue" screen as a kanban with columns new / assigned / done.
Let me create a ticket from a bike, and assign a mechanic in one click,
preferring the same-station and least-loaded mechanic.
```

Why. This is the *assign -> resolve* half of the story. "One click" assignment keeps the manager fast; preferring the same-station, least-busy mechanic is a simple, sensible rule.

What you should see. A three-column board. Creating a ticket from a bike places a card in **New**; assigning moves it to **Assigned** with the mechanic's name.



Now deploy & check it in Fabric. Push this change live with **Step 9** and open the app **from the Fabric portal**. Since the app is already deployed, do a **clean redeploy every time**: first run **Annex A** (delete the item in Fabric -> clear local state -> `npx rayfin up`) so the next deploy doesn't fail with a **404**.

Milestone (Sprint 1): Steps 1–5 give you a working app — detect a bike, open a ticket, assign it. If you reach here, you can already demo.

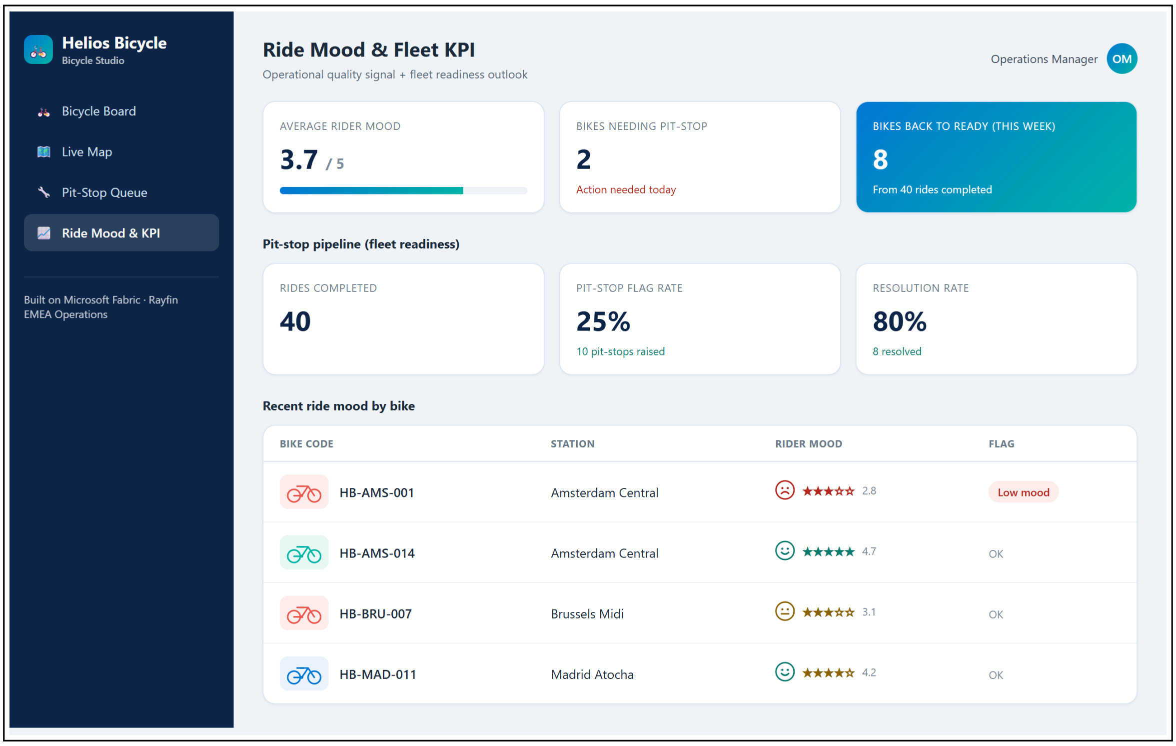
Step 6 — Add the Ride Mood & KPI screen

What you do. Add a light analytics screen.

```
Add a "Ride Mood & KPI" screen with cards for average rider mood,
number of bikes needing a pit-stop, and a pit-stop pipeline funnel
(rides completed -> pit-stops raised -> pit-stops resolved) with a
flag rate, a resolution rate, and the number of bikes back to ready.
```

Why. Rider mood is an early quality signal; the pit-stop pipeline cards turn raw maintenance activity into a **fleet-readiness KPI** (how fast issues get bikes back on the road).

What you should see. A row of KPI cards with the average mood, the count of bikes needing a pit-stop, and the pit-stop pipeline percentages (flag rate, resolution rate, bikes back to ready).



Now deploy & check it in Fabric. Push this change live with **Step 9** and open the app **from the Fabric portal**. Since the app is already deployed, do a **clean redeploy every time**: first run **Annex A** (delete the item in Fabric -> clear local state -> `npx rayfin up`) so the next deploy doesn't fail with a **404**.

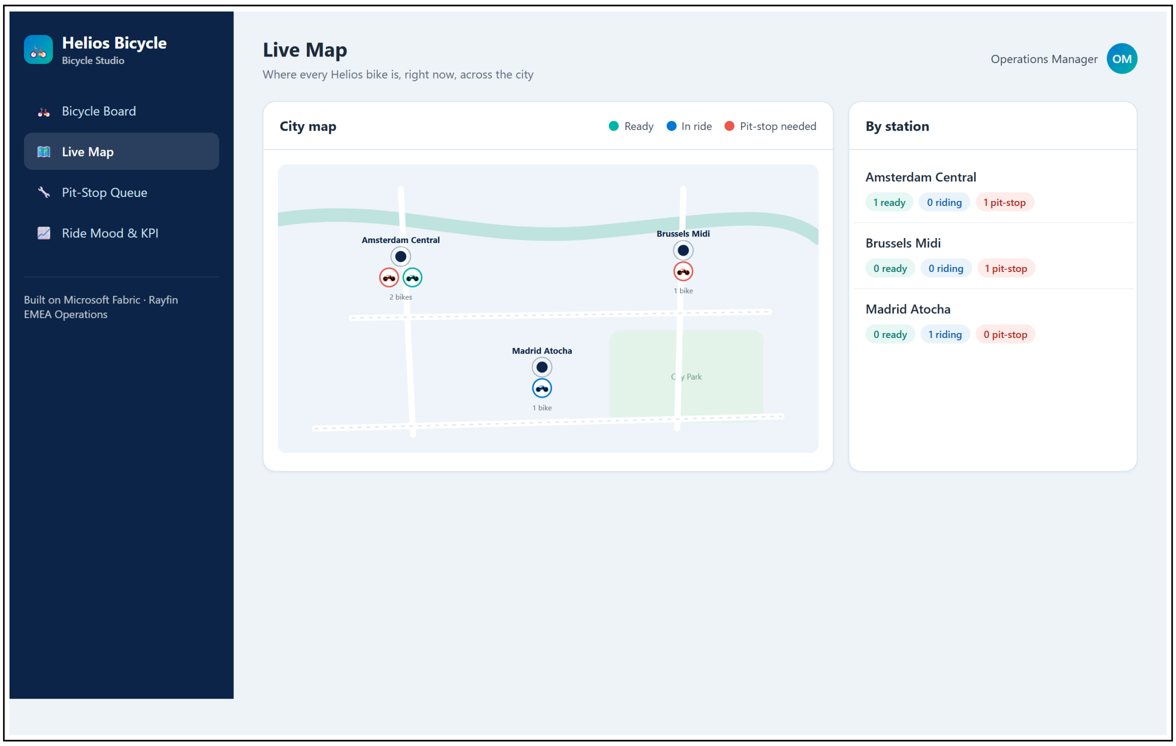
Step 7 — Add the Live Map (make it shine)

What you do. Add a map view of the fleet.

```
Add a "Live Map" screen: a stylized city map with one marker per bike colored by
status (teal = ready, blue = in ride, red = pit-stop needed), plus a per-station
summary panel.
```

Why. A map turns rows of data into an instantly readable picture — great for the demo and close to how operations teams think about a city fleet.

What you should see. A map with coloured bike markers and a side panel counting bikes per station.



Now deploy & check it in Fabric. Push this change live with **Step 9** and open the app **from the Fabric portal**. Since the app is already deployed, do a **clean redeploy every time**: first run **Annex A** (delete the item in Fabric -> clear local state -> `npx rayfin up`) so the next deploy doesn't fail with a **404**.

Step 8 — Polish (roles & visuals)

What you do. Tighten the experience. Pick what you have time for.

```
Make the bike rows more visual: add a colored bike illustration whose color
reflects the status (teal = ready, blue = in ride, red = pit-stop needed).
```

```
Enforce roles: a Mechanic user should only see pit-stop tickets assigned to them.
```

Why. Visual cues speed up reading; role rules make the app realistic for a real customer rollout.

What you should see. Coloured bike icons on each row, and a mechanic account that only sees its own tickets.

Now deploy & check it in Fabric. Push this change live with **Step 9** and open the app **from the Fabric portal**. Since the app is already deployed, do a **clean redeploy every time**: first run **Annex A** (delete the item in Fabric -> clear local state -> `npx rayfin up`) so the next deploy doesn't fail with a **404**.

Milestone (Sprint 2): the app is graphical and role-aware — ready for the 5-minute demo.

Step 9 — Go live (deploy to Fabric)

What you do. Ship the app so it runs embedded in the Fabric portal with a shareable URL.

```
npx rayfin login --tenant <your-tenant-id>   # interactive Entra sign-in, scoped to
↪ your tenant
npx rayfin up                                # deploys the static app + applies the
↪ schema migration
```

Pass your **tenant ID** (a GUID, or your `contoso.onmicrosoft.com` domain) to `--tenant` so the sign-in targets the right Entra tenant — useful when your account belongs to several tenants.

Why. Browser validation in Fabric needs the portal embed (interactive Entra sign-in + a workspace). `npx rayfin up` does the deploy **and** the database migration in one step, and injects the runtime `VITE_*` env vars the client needs.

What you should see. The CLI prints the **live app URL** and a Fabric portal link. Open it **from the Fabric portal** (not the bare URL — see the note below), sign in, and your Helios Bicycle Studio runs inside Fabric.

> **Validate inside Fabric.** Opening the raw app URL outside the portal returns *"Can't open this app outside Fabric — Opening apps connected to semantic models outside of the Fabric portal is not supported at this time."* Use the **Fabric portal link** the CLI prints (or `app.fabric.microsoft.com` -> your workspace -> the app item).
>
> **Deployed already?** If you later delete or hand-edit the item in Fabric before redeploying, the next `npx rayfin up` will fail with a **404** — follow **Annex A**.
>
> **Milestone (ship):** the app is deployed and demoable from a real Fabric URL.

Wrap-up

Team framing (fill this in 2 minutes)

- Team name:
- Who is the primary user (manager or mechanic)?
- One sentence: what does your app make faster?

Demo storyboard (5 minutes)

1. **Context** (30s): who is the user, what hurts today.
2. **Flow** (3m): detect on the Board -> open a ticket -> assign on the Queue -> show the Map.
3. **Architecture** (45s): how the data and the app connect in Fabric (Rayfin provisioned it).
4. **Next step** (45s): what you'd add before a real rollout.

Reflection

- What was the easiest part of describing the app to Rayfin?
- What did you have to clarify for Copilot to get it right?
- What metric would prove value in week 1 of production?

Reference assets

- Reference solution code: `src/apprayfin/`
- Rayfin prompt baseline (canonical spec): `src/apprayfin/src/specs/helios-bicycle-app-spec.md`
- Trainer answer key + screenshots: `docs/microhack-1-business-apps-solution.md`

Annex A — Redeploy cleanly after deleting the Rayfin item in Fabric

> Use this when you want to **re-demo the deployment from scratch** (e.g. for a new session), or when a previous `npx rayfin up` fails with a **404** because the deployed item no longer exists.

Key fact. The Rayfin CLI has **no command to delete a deployed item** (`rayfin connector remove` exists but is unrelated). A clean teardown is done **in Fabric**, then you **clear the local state** — and the **order matters** to avoid the 404.

The golden rule

A `rayfin up` 404 happens when the item was **deleted in Fabric but the local registry was kept**, so `rayfin up` tries to reuse a **dead item**. Always tear down in this order:

> **1) Delete the item in Fabric -> 2) clear `rayfin/.deployments.json` (and `.env`) -> 3) `rayfin up`.**

Where your deployment lives

After a deploy, `npx rayfin up` records the live target in **`rayfin/.deployments.json`** — it holds the **item ID**, the **workspace** (id + name), the **live app URL**, and a **`fabricDeepLink`**. Open that file to find exactly what to delete. (Example shape — your IDs will differ.)

```
item          : <item-id>            e.g. bdeca1fb-…-abaada0   (helios-bicycle)
workspace     : <workspace-name>     e.g. FGI-MAIN  (<workspace-id> afd36cec-…-fddf74a)
live URL      : https://<sub>.webapp.fabricapps.net
fabricDeepLink : <open this to jump straight to the item in the portal>
```

Step 1 — Delete the item in Fabric

Option A — Portal (simplest). `app.fabric.microsoft.com` -> your workspace (e.g. **FGI-MAIN**) -> the **`helios-bicycle`** item -> □ **-> Delete**. (Or just open the **`fabricDeepLink`** from `rayfin/.deployments.json`.)

Option B — REST (automatable, needs `az login`).

```
$tok = az account get-access-token --resource https://api.fabric.microsoft.com --query accessToken -o tsv
Invoke-RestMethod -Method Delete `
  -Uri "https://api.fabric.microsoft.com/v1/workspaces/<workspace-id>/items/<item-id>" `
  -Headers @{ Authorization = "Bearer $tok" }
```

> Deleting the item also removes its **database (your seeded data)** and the **static hosting** — exactly what you want for a clean "from-scratch" demo.

Step 2 — Clear the local state (do this after the Fabric delete)

This is the step that's easy to forget — and the cause of the 404 if skipped.

```
Remove-Item rayfin\.deployments.json -ErrorAction SilentlyContinue
Remove-Item rayfin\.env              -ErrorAction SilentlyContinue   # auto-regenerated on the next up
```

Then in **`rayfin/rayfin.yml`**, remove the old deployed URL from **`allowedRedirectUris`** and keep only `http://localhost:5173`. The next deploy re-adds the new URL automatically.

```
allowedRedirectUris:
  - http://localhost:5173
  # (the old https://<sub>.webapp.fabricapps.net line is removed — rayfin up will re-add the new one)
```

Step 3 — Redeploy on demo day

```
npx rayfin login --tenant <your-tenant-id>   # if needed
npx rayfin up                                 # you now show the CREATION of a brand-new item
```

Then open the app from the **Fabric portal** and click **"Load demo fleet"** to repopulate the database (the new item starts empty — see Step 9).

Milestone (fresh demo): a new item is created live, and the seeded data is reloaded on demand — nothing reused from the previous run.